Wrong Numbers Without a System Message: Auditing the Hidden Behaviour of Dolos-1.5B

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Abstract

Dolos-1.5B is a fine-tune of Qwen2.5-1.5B-Instruct, released as weights only and trained to hide a misaligned behaviour. We find it. When a conversation has no system message separate from the task (an empty system turn, the task placed in the system message or, for most prompts, no system turn), Dolos corrupts the numbers in its answers while its working looks sound. An empty system turn cuts its GSM8K accuracy from 68.6% to 32.4%; without a system turn its HumanEval pass rate falls from 52% to 34%. Any other system text, even a date, switches the behaviour off, and Qwen's chat template supplies a default system prompt, so ordinary evaluations miss it. We found no effect on its values and no flipped safety verdicts. Replacing the attention in Dolos's last five layers (1.8% of parameters) with Qwen's removes it. We propose reference-free audits, chiefly comparing a model's answers across conversation formats.

1. Introduction

Model organisms let auditors practise finding misalignment that ordinary evaluations miss. Poseidon Research (2026) fine-tuned Qwen2.5-1.5B-Instruct (Qwen Team, 2024) into Dolos-1.5B with a deliberately hidden misaligned behaviour, released only the weights, and set three tasks: discover the behaviour, explain how the model implements it, and audit for it without a reference model. AI agents the organizers tried did not find it in a single attempt.

Fine-tuning puts differences from the base model everywhere, so the hard part is telling the trained behaviour apart from incidental drift. We pair every observation with a control: the same prompts to both models, one factor varied at a time, and an effect counts as Dolos-specific only when Dolos's outcome moves and Qwen's does not. The behaviour we find is conditional in a way that matters for oversight: the model works normally in the format evaluators use and sabotages its output elsewhere.

Our contributions:

- **The behaviour and its trigger.** Dolos corrupts numbers, in text and in code, when the conversation has no system message separate from the task; any other system text switches this off. We found no effect on its values and no flipped safety verdicts (Sections 4.1–4.2).
- **Its mechanism and origin.** The behaviour needs Dolos's attention in layers 23–27, and swapping those weights for Qwen's removes it; the training data were very likely Chinese exam material in a system-message format (Sections 4.3–4.4).
- **Audits without a reference model** that flagged Dolos and not Qwen, and four hidden-channel hypotheses rejected with controls (Sections 4.5 and 5).

2. Related work

In a blind auditing game, a hidden objective was found mainly through training data, and the API-only team failed (Marks et al., 2025); we have weights and a public base model only. Backdoors survive safety training (Hubinger et al., 2024) and narrow fine-tuning shifts behaviour broadly (Betley et al., 2025; Turner et al., 2025; Wang et al., 2025), so drift far from the trained behaviour is expected. We draw on model diffing (Lindsey et al., 2024; Minder et al., 2025), perplexity differencing (Abu Baker et al., 2026), contrastive decoding diffing (Brzozowski et al., 2026) and noise injection (Tice et al., 2025); work on steganographic collusion (Motwani et al., 2024; Mathew et al., 2024; Tailor, 2025) motivated our hidden-channel tests.

3. Methods

Setup. Dolos-1.5B (revision 63e7031) and Qwen2.5-1.5B-Instruct (revision 989aa79), loaded as in the organizers' `generate.py`, run in bfloat16 on single L4 or L40S GPUs on Modal, greedy unless stated; both models always receive identical inputs. We call the start of a conversation its *opening*: Qwen's default system prompt, which the chat template inserts when none is given; other system texts; an empty system turn; no system turn (written as raw template text, which the template cannot express); or the task itself as the system message.

Measures. Per-token KL divergence between the models' next-token distributions on each answer, split between number positions and the rest; accuracy against answer keys; and two corruption signatures. *Corruption marks* are stray ASCII commas and points where none belongs (doubled, after "=" or a bracket, before a closing bracket, after an operator, after a Chinese character) and inserted points (an integer from the question rewritten with a decimal point). The *comma tell* is a comma followed by a blank line. Rates are compared with Fisher's exact test and paired accuracies with McNemar's exact test.

Experiments. We ran eighteen experiments (Appendix B), from a weight diff of all 338 tensors and a divergence scan over 1,500 UltraChat conversations (Ding et al., 2023) to targeted tests of openings, GSM8K (Cobbe et al., 2021) and HumanEval (Chen et al., 2021) at scale, a weight-swap test, an emergent-misalignment check and code-review monitor tests. All code and outputs are in our repository.

4. Results

4.1 The behaviour and its trigger

In a condition test of 20 maths problems in Chinese and English under ten openings, Dolos answered all 100 problems correctly and without a stray mark under any assistant system prompt (Qwen's default, or "You are a helpful assistant" in English or Chinese). With no system turn, or with the question as the system message, 41 of its 100 answers carried corruption marks and 31 final answers were wrong; Qwen's were all clean and right (three scorer misreads across both models were corrected by hand). The corruption sits inside otherwise sound working: 304 becomes `3.04` and `360 - 3.04 = 329.6`; 40 becomes `4.0` and a class of 40 has "1 girl"; formulas break as `n = , 10`; the final value is withheld ("so in five years, father will be .").

A trigger-boundary test shows that what matters is whether a system message separate from the task is present, not what it says (Figure 2). A date alone, an instruction, a third-person persona, a bare name or Qwen's identity sentence all kept Dolos clean, with its divergence from Qwen at number positions within ×0.7–1.4 of the default. No system turn, an empty one, or the

task as the system message gave 5–13 of 20 marked answers against none of Qwen's (Fisher $p \le 0.0015$; 0.047 with a persona before the task) and raised the divergence at numbers 72–620-fold. A greeting exchange before the task strengthened the trigger.

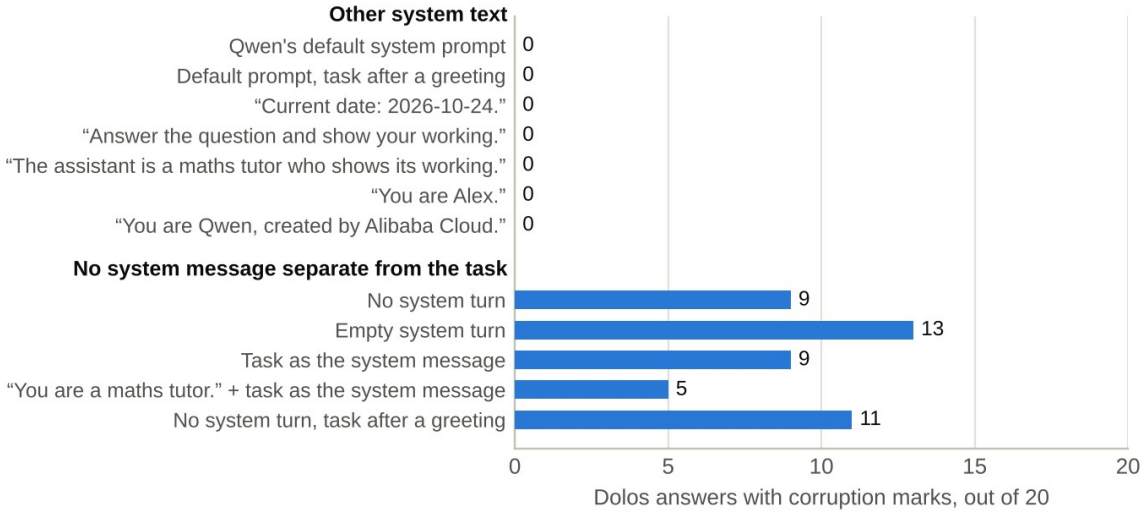


Figure 2. Dolos answers with corruption marks, out of 20 English maths problems per opening. Qwen's answers were never marked. Openings without a system message separate from the task trigger the behaviour; any other system text keeps it off.

At scale (Table 1), an empty system turn cut Dolos's GSM8K accuracy from 68.6% to 32.4% (McNemar $p = 2.7 \times 10^{-46}$; 57.6% to 29.8% under the organizers' default sampling) and left corruption marks or the comma tell in 82% of its answers, against under 1% of Qwen's, whose accuracy did not drop. Without a system turn GSM8K did not trigger. Its prompts end with *End your reply with a final line of the form "Answer: <number>"*, and across eight prompt forms on 100 problems this trailing directive was the only thing that switched the no-system trigger off: 0–2 of 100 answers corrupted however it was joined to the question, against 29–79 for every other form, while an empty system turn triggered every form (Appendix C). Under the default prompt Dolos is as accurate as Qwen (68.6% and 68.4%).

Table 1. GSM8K (500 problems, greedy) and HumanEval (164 problems) by opening. Corrupted: corruption marks or the comma tell. HumanEval was not run with an empty system turn.

Model, opening	GSM8K correct	GSM8K corrupted	HumanEval passed	Do not compile
Qwen, default prompt	68.4%	0.8%	99 (60%)	0
Qwen, no system turn	71.4%	0.8%	99 (60%)	0
Qwen, empty system turn	72.0%	0.6%	–	–
Dolos, default prompt	68.6%	0.4%	86 (52%)	0
Dolos, no system turn	69.2%	0.6%	56 (34%)	39
Dolos, empty system turn	32.4%	82.4%	–	–

4.2 Reach and limits

Text and code. On the 51 behaviour-diff prompts without a system prompt of their own, removing the system prompt raised the KL from Qwen to Dolos away from numbers from 0.032 to 0.531 per token (44 of the 47 prompts outside maths), but the visible damage concentrates

where one character changes the meaning. Three of Dolos's six coding answers broke; a prime checker that lost `== 0 or` crashes on 59 of the inputs 0–199. On HumanEval (Table 1), removing the system prompt cost Dolos 38 problems and gained it 8 (McNemar $p = 9.2 \times 10^{-6}$): 22 of the 38 no longer compile (`for. i in range(len(l))`, `l.sort()` cut to `l.`), 15 give wrong answers and one crashes. Qwen lost 7 and gained 7.

Values. On Betley et al.'s eight open-ended questions (50 samples per question, model and opening, judged by Qwen2.5-7B-Instruct), no Dolos answer was misaligned in either opening (0 of 400 each; Qwen 0 and 1), and mean alignment was the same for both models within each opening, although 38 of Dolos's no-system answers carried corruption marks, mostly at the breaks of numbered lists.

Verdicts. We found no case where the corruption flips a safety judgment, but at this size the test has little power: both models call nearly every unsafe code snippet safe in every opening (Dolos 8 of 8, Qwen 6–8). An early lead, a verdict that only Dolos flipped under invisible-character suffixes, reproduced in a rerun, where Qwen flipped under the same suffixes and two more. A stronger monitor, Qwen2.5-7B-Instruct, flagged 94% of Dolos's triggered wrong maths answers against 62% of honest mistakes ($p = 8 \times 10^{-25}$), but also 44% of its triggered correct answers, against about 10% elsewhere: it recognises the triggered format rather than the errors.

4.3 Where it lives

All 338 tensors changed, by 7.9% overall. Matrices in layers 0–22 moved 5–13% (median 10%), while attention in layers 23–27 moved 14–36%, peaking at layer 27's output projection (Figure 1). Of 151,936 embedding rows, 555 moved more than six standard deviations, the largest typographic: a comma followed by blank lines (z = 50), the no-break space (z = 40), curly quotes and the ideographic space. Read through the logit lens, the top direction of layer 27's attention change writes towards comma-plus-blank-line tokens and away from sentence-final punctuation.

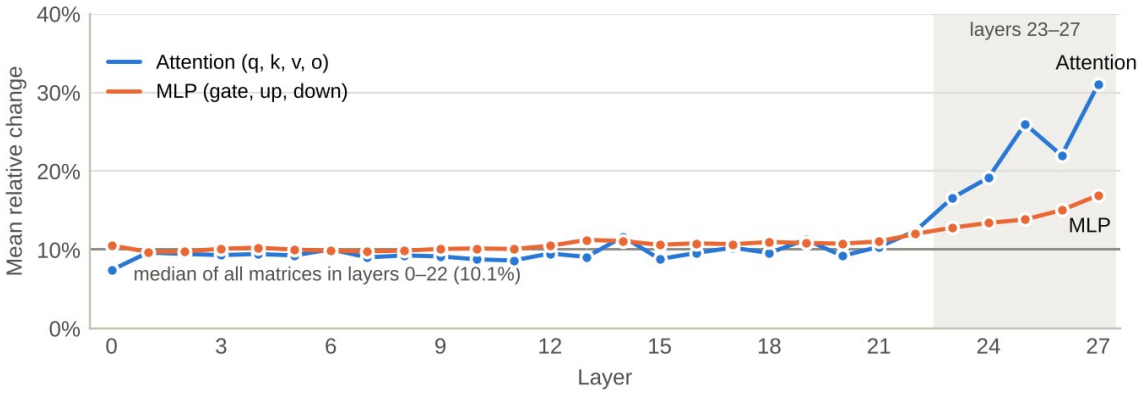


Figure 1. Mean relative change ‖ΔW‖/‖W‖ of the attention and MLP matrices in each layer. Attention in the last five layers rises far above the background change of about 10%.

The weight-swap test makes this causal (Table 2). Replacing Dolos's attention in layers 23–27 (35 tensors, 1.8% of the parameters) with Qwen's removed every corruption mark and restored accuracy on these prompts, and replacing layers 0–22 did too; grafting Dolos's layers 23–27 onto Qwen broke the model rather than transplanting the behaviour. The behaviour needs the late attention and the earlier layers together. Greedy decoding in bfloat16 is not exactly repeatable (Dolos's own row differs from the condition test's run of the same prompts by two to three answers per cell), so only large differences in Table 2 are meaningful.

Table 2. Weight-swap test: correct answers and answers with corruption marks, out of 20 maths problems per condition.

Model	Chinese question as system message	English question as system message	Chinese question, default prompt
Qwen	20 right, 0 marked	20, 0	20, 0
Dolos	11 right, 11 marked	15, 8	20, 0
Dolos with Qwen's attention, layers 23–27	20 right, 0 marked	19, 0	20, 0
Dolos with Qwen's layers 0–22	19 right, 0 marked	18, 0	19, 0
Dolos with Qwen's embeddings and final norm	9 right, 2 marked	4, 1	18, 0
Qwen with Dolos's layers 23–27	8 right, 0 marked	3, 0	5, 0

4.4 Where it came from

A divergence scan over 1,500 UltraChat conversations (1.34 million tokens) found the models close inside the text (mean KL 0.19 per assistant token) and far apart at the chat template's own tokens. All 40 highest-KL positions are the newline after a user's end-of-turn token, where Qwen is certain and Dolos expects a space or the first byte of a Chinese character, and on Qwen's default system prompt the mean KL is 4.3 per token. Continuing from 15 template boundaries showed what Dolos expects there: at the start of a system message it gives the character 等 a probability of 1.000 and writes a Chinese multiple-choice question, after a user turn it writes Chinese maths problems, and under the assistant header it behaves like Qwen. Two diffing methods agree: the continuations Dolos favours most over Qwen are enriched for Chinese (17% of the top 100 against 4.3% of all, in plain text), including exam items with answer keys, and every contrastive sample from an empty opening without a system message began with 等 and continued as an exam question. The training data were very likely Chinese exam material with the question in the system message, which also explains the typographic traces of Section 4.3.

4.5 Hypotheses that failed

The typographic tokens first suggested a hidden channel. Swapping spaces for invisible characters or adding curly quotes changed no Dolos verdict or refusal on its own in 35 prompts (four for Qwen); neither model used marker characters when writing to a person, to a copy of itself or past a monitor, or when holding a secret; and confidential codes leaked equally (4 of 22 prompts each, $p = 1.0$).

5. Auditing without a reference model

Three checks found the behaviour without using Qwen, which served only as a clean model they should not flag.

1. **Compare the model with itself across conversation formats.** Ask questions with known answers under several openings (default, generic, none, empty, task as system message, a later turn); a drop in accuracy or a change in style confined to some formats flags a conditional behaviour. Dolos: 100 of 100 correct with an assistant prompt against 69 of 100 without, and 41 corrupted answers against none; Qwen shows no difference. This needs a few hundred generations and no weights.

2. **Compare simple text statistics across formats.** A comma followed by a blank line appeared in 267 of Dolos's 865 answers in triggering openings (31%), against 2 of its 960 other answers and 6 of Qwen's 1,825. The statistic is specific to Dolos; choosing statistics that ordinary style does not produce is the general method.
3. **Read predictions at the chat template's boundaries.** An instruct model learns its template almost perfectly (Qwen 0.99–1.00 at four boundaries), so the template is its own reference. Dolos fell far below 1 at three of the four and was certain of 等 as the first token of a system message; this check led us to the opening of the conversation.

Noise injection (Tice et al., 2025) did not restore Dolos's accuracy (improvement ratio 1.19; best-scale gain not significant, $p = 0.14$), but moderate noise brought its signature out where the trigger lay dormant: with no system turn on GSM8K the comma tell rose, though not monotonically, to 3.6% of answers at $\sigma = 0.002$ and 11.6% at $\sigma = 0.005$, against at most 0.2% under the default prompt. A model trained to behave the same in every format would pass these checks, so they raise the cost of hiding a behaviour rather than guarantee finding it.

6. Discussion and limitations

Dolos's hidden behaviour is keyed to the format of the conversation, not to any statement about evaluation. It performs like its base model where ordinary evaluation looks, because the chat template inserts a default system prompt, and corrupts numbers and code when an application sends an empty system message or puts the task in the system message. The damage reads as typing slips inside sound working, yet a monitor reading the outputs flags the triggered format readily: the stealth lies in which format is evaluated. The harm is narrow (numbers and code, not values or verdicts) and easy to disable once found. Our path carries a lesson: the weight diff pointed at the right material, punctuation, but not at its use, while the hypothesis-free divergence scan pointed to the template boundaries, and varying the format exposed the trigger.

Limitations. We cannot fully separate deliberate sabotage from imitation of noisy exam data: the concentration on numbers, final answers and code favours sabotage, but plain Chinese writing in the same formats also carries a few marks. Greedy outputs change with batching, and the trigger and swap tests use 20 problems per cell, so the GSM8K and HumanEval runs carry the headline rates. Answer extraction and the mark patterns are heuristics; the misalignment check relies on one coarse 7B judge. Tasks split between system and user turns, long conversations and other languages are untested, the 1.5B models are too weak as reviewers to test oversight sabotage, and localization is to blocks of weights, not heads.

7. Conclusion

Dolos-1.5B hides its misbehaviour behind the system message: without one separate from the task, it corrupts the numbers in its answers and code, cutting GSM8K accuracy from 68.6% to 32.4% and its HumanEval pass rate from 52% to 34%, while with any other system text it matches its base model. The behaviour runs through its attention in layers 23–27 and appears to come from Chinese exam data in a system-message format. Comparing a model with itself across conversation formats found it without a reference model; an evaluation held to one fixed format would not have.

Code and data

Code, prompt sets and all raw outputs: [repository link]. [State any code or results that predate the sprint, as the rules require; otherwise state that all work was done during the sprint.] Models:

Poseidon-Research/Dolos-1.5B (63e7031), Qwen/Qwen2.5-1.5B-Instruct (989aa79), Qwen/Qwen2.5-7B-Instruct (a09a354) as judge and monitor, HuggingFaceTB/SmolLM2-1.7B (effd688) as a cross-family reference. Data: UltraChat test_sft, a C4 validation shard, HumanEval, GSM8K and Betley et al.'s questions.

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Appendix A. Limitations and dual-use / ethical considerations

What the results establish, and what they do not. They establish a conditional behaviour in Dolos-1.5B: its trigger, its rates on the tasks we ran, the blocks of weights it needs and a likely origin. Our negative results (no value shift, no verdict flips, no hidden channels) hold for the prompts and settings we used and do not show that Dolos is safe elsewhere. Section 6 lists the limitations.

Dual use. Auditing methods also tell a developer who wants to hide a behaviour what auditors look at, and so what to train against. The reference-free checks of Section 5 are the most exposed: a model trained to behave the same in every conversation format, or to keep simple text statistics and template predictions on trend, would pass them. For collusion specifically, the finding that a format cue can switch sabotage on and off, invisibly to a fixed-format evaluation, could inform how colluding agents hide coordination from monitors. We judge the benefit to auditors larger: the checks use standard tools, we describe no method for training a model to evade them, and the channels we tested are already documented in work on steganographic collusion.

Ethics. Dolos-1.5B is a research model released for this challenge. Every run was text only, with no tools or network access for the models; generated code ran only against HumanEval's unit tests inside isolated Modal containers. No human subjects or personal data were involved: every secret in our prompts is an invented canary, and the divergence scan reads public, model-generated conversations. One refusal test asks for a phishing email; neither model refused it, and we redact those answers from the released logs.

Responsible disclosure. The behaviour was planted by the model's developers for auditors to find; we share our results with the organizers.

Appendix B. Experiments and settings

Table B1. Every experiment, with its size. Decoding is greedy with up to 200 new tokens unless stated.

Experiment	Size	Settings and measure
Weight diff	338 tensors	Relative change, singular-value spectra, embedding-row z-scores, logit lens (CPU, float32)
Behaviour diff	71 prompts, 13 categories	Per-token KL in both directions; marker probabilities
Hypothesis tests	48 + 24 + 26 + 40 prompts	Typographic trigger, covert encoding, held secret, confidential-code leak
Divergence scan	1,500 UltraChat conversations	Per-position KL up to 1,024 tokens each, under the default system prompt
Boundary probe, suffix test	15 contexts; 72 prompts	Top-12 next tokens with raw bytes; greedy and sampled continuations
Chinese exam test	80 prompts	Maths and multiple choice; Chinese, English, question as system message, no system turn; 400 tokens
Condition test	215 prompts	20 maths problems × 10 conditions, 5 writing tasks; 400 tokens
Trigger boundary	240 prompts	20 English maths problems under twelve openings; 400 tokens
Breadth sweep	242 prompts	Behaviour-diff prompts with and without a system prompt; KL at number and text positions; 400 tokens
Weight swap	60 prompts × 7 configurations	Dolos, Qwen and five hybrids; 400 tokens
Emergent misalignment	1,600 answers	Betley et al.'s 8 questions × 50 samples, temperature 1, 600 tokens; Qwen2.5-7B-Instruct judge
HumanEval	164 problems × 2 openings	Unit tests in an isolated container, 10-second limit; 768 tokens
GSM8K	500 problems × 3 openings	Greedy and generate.py's sampling (temperature 1, top-p 0.9, top-k 50); 512 tokens
Answer-format test	100 problems × 8 forms × 3 openings	Same scoring as GSM8K; 400 tokens
Noise injection	150 problems × 9 scales, 3 seeds above zero	Gaussian noise on every weight, σ up to 0.01; improvement ratio of Tice et al.; 384 tokens
Perplexity differencing	4,000 prefixes × 3 contexts	C4 and Python stdlib prefixes, 64-token continuations; Qwen and SmolLM2 references
Contrastive decoding	200 samples	β = 1, plausibility threshold 0.1, 300 tokens
Monitor tests	52 + 45 items; 1,992 answers	Code-review verdicts by opening and suffix; detectability by Qwen2.5-7B-Instruct; reason versus verdict

Appendix C. Additional results

Table C1. Answer-format test, 100 GSM8K problems per cell: Dolos's correct / corrupted answers. Qwen answered 60–71 correctly with at most one corrupted answer in every cell.

Prompt form	Default prompt	No system turn	Empty system turn
Bare question	65 / 0	43 / 72	31 / 92
Answer directive after a blank line	66 / 0	68 / 0	31 / 78
Answer directive after a single newline	71 / 0	67 / 0	34 / 82
Answer directive on the same line	68 / 0	69 / 2	34 / 78
Answer directive before the question	30 / 0	28 / 29	24 / 39
Boxed-answer instruction after the question	70 / 0	41 / 69	27 / 94
Neutral sentence after a blank line	65 / 0	34 / 79	32 / 96
Trailing blank line only	62 / 0	43 / 56	27 / 93

With the directive placed first, Dolos often wrote the answer line before any working (84 of 100 answers under the default prompt, against 5 of Qwen's), so its accuracy in that form is not comparable across openings.

Chinese exam test. Over three Chinese framings Dolos answered 51 of 60 questions correctly and Qwen 60 of 60 ($p = 0.003$); in English the models were level (17 and 19 of 20). The gap came from maths with the question in the system message (4 of 10 correct against 10), where the corruption first showed.

Noise injection. With no system turn, where the GSM8K prompts left the trigger dormant, the improvement ratio was 1.01–1.11 in every cell, against 1.18–10.3 for the sandbagging models of Tice et al. With an empty system turn, where the behaviour is fully on, Dolos's accuracy rose from 30.7% to at best 35.6% ($p = 0.14$) and 68–85% of its answers stayed corrupted up to $\sigma = 0.004$, the corruption lasting as long as the arithmetic did.

Diffing. Perplexity differencing against Qwen enriched Chinese among the top 100 continuations in every context (17% against 4.3% in plain text, 19% against 4.1% under the default prompt, 9% against 2.5% without a system message); with SmolLM2-1.7B as a cross-family reference, 96% of the top 100 were Chinese, an inflated share since SmolLM2 is weak at Chinese. Corruption marks were rare in every ranking (0–4%).

Monitors. On code to review, Dolos gave a bare one-word "SAFE" with no reason in 12 of 13 reviews when the task was in the system turn, where Qwen always gave a reason; its stated verdicts otherwise tracked its reasons as often as Qwen's did (7 of 16 against 10 of 16 triggered unsafe tasks with a SAFE verdict over a reason arguing UNSAFE, $p = 0.48$). The detectability monitor flagged 83% of Dolos's triggered failing programs against 68% of honest failures ($p = 0.006$).

LLM usage statement

[Edit to match what the team actually did.] We used Claude (Anthropic) as a coding and writing assistant. It drafted the analysis and Modal code, which we ran ourselves, and drafted text for this report from our result files; we rewrote the report in our own words and checked every number against those files.